THE POLARIZED STACK IS LIMIT PRESERVING

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Definition artin-definition-limit-preserving (canonical tag 07XL). cited at statement line 4287. The property being established.
- Lemma limits-lemma-descend-finite-presentation (canonical tag 01ZM). cited at proof line 4300. Schemes colimit.
- Lemma limits-lemma-descend-modules-finite-presentation (canonical tag 01ZR). cited at proof line 4305. Modules colimit.
- Lemma limits-lemma-descend-invertible-modules (canonical tag 0B8W). cited at proof line 4306. Invertible modules colimit.
- Lemma limits-lemma-eventually-proper (canonical tag 081F). cited at proof line 4309. Eventual properness.
- Lemma limits-lemma-descend-flat-finite-presentation (canonical tag 04AI). cited at proof line 4310. Eventual flatness.
- Lemma limits-lemma-limit-ample (canonical tag 09MT). cited at proof line 4314. Eventual ampleness.

Source reference. Original definition 07XL: sources/stacks-chow/tags.txt, source lines 10081–10081. Cited at statement line: the property being established; load-bearing.

Source reference. Original lemma 01ZM: sources/stacks-chow/tags.txt, source lines 2599–2599. Cited at proof line 4300: schemes colimit.

Source reference. Original lemma 01ZR: sources/stacks-chow/tags.txt, source lines 2604–2604. Cited at proof line 4305: modules colimit.

Source reference. Original lemma 0B8W: sources/stacks-chow/tags.txt, source lines 14154–14154. Cited at proof line 4306: invertible modules colimit.

Source reference. Original lemma 081F: sources/stacks-chow/tags.txt, source lines 10215–10215. Cited at proof line 4309: eventual properness.

Source reference. Original lemma 04AI: sources/stacks-chow/tags.txt, source lines 5617–5617. Cited at proof line 4310: eventual flatness.

Source reference. Original lemma 09MT: sources/stacks-chow/tags.txt, source lines 12191–12191. Cited at proof line 4314: eventual ampleness.

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Lemma 0.2. *The stack in groupoids $\mathcal{P}olarized$ is limit preserving (Artin's Axioms, Definition 07XL).*

Proof. Let I be a directed set and let $(A_i, \varphi_{ii'})$ be a system of rings over I. Set $S = \mathrm{Spec}(A)$ and $S_i = \mathrm{Spec}(A_i)$. We have to show that on fibre categories we have

$$\mathcal{P}olarized_S = \mathrm{colim}\, \mathcal{P}olarized_{S_i}$$

We know that the category of schemes of finite presentation over S is the colimit of the category of schemes of finite presentation over S_i, see Limits, Lemma 01ZM. Moreover, given $X_i \to S_i$ of finite presentation, with limit $X \to S$, then the category of invertible $\mathcal{O}_X$-modules $\mathcal{L}$ is the colimit of the categories of invertible $\mathcal{O}_{X_i}$-modules $\mathcal{L}_i$, see Limits, Lemma 01ZR and 0B8W. If $X \to S$ is proper and flat, then for sufficiently large i the morphism $X_i \to S_i$ is proper and flat too, see Limits, Lemmas 081F and 04AI. Finally, if $\mathcal{L}$ is ample on X then $\mathcal{L}_i$ is ample on X_i for i sufficiently large, see Limits, Lemma 09MT. Putting everything together finishes the proof. $\square$